

ifemm

Output Class Attributes

Version 0.1-alpha — August 18, 2026

Attributes

Top-Level

Attribute	Type	Description
format_version	float	Format version of the file.
frequency_hz	float	Frequency for harmonic problems.
solver_precision	float	Computing precision within FEMM.
min_angle_deg	float	Angle restriction for the triangulation.
model_depth	float	Typical length in z-direction (for 2D planar).
length_unit	str	Unit used for lengths (e.g., 'millimeters', 'inches').
problem_type	str	The 2D problem is either planar or axisymmetric.
coordinate_system	str	Definition of the coordinate system.
comment_text	str	Problem description.
vector_x	list[float]	X-coordinates of the solution points.
vector_y	list[float]	Y-coordinates of the solution points.
vector_a	list[float]	Magnetic vector potential values at each point.

Materials

Attribute	Type	Description
name	str	Name of the material.
permeability	tuple[float, float]	Permeability in x and y directions.
coercive_force	float	Coercive force in A/m.
coercive_angle	float	Angle of coercive force in degrees.
current_density	tuple[float, float]	Real and imaginary current density.
conductivity	float	Electrical conductivity in S/m.
lamination_thickness	float	Lamination thickness in mm.
hysteresis_angle	float	Hysteresis angle for AC analysis.
hysteresis_angle_x	float	Hysteresis angle in x-direction.
hysteresis_angle_y	float	Hysteresis angle in y-direction.
lamination_type	int	Type of lamination.
lamination_fill	float	Lamination fill factor.
num_strands	int	Number of strands in a stranded conductor.
wire_diameter	float	Wire diameter in mm.
bh_curve	list[tuple]	B-H curve data points (H, B).

Boundaries

Attribute	Type	Description
name	str	Name of the boundary condition.
boundary_type	int	Boundary type
prescribed_a	float	Prescribed A value.
prescribed_a_x	float	Prescribed A x-component.
prescribed_a_y	float	Prescribed A y-component.
phase_angle	float	Phase angle in degrees for AC analysis (Phi).
mixed_c0	float	Mixed boundary C0 coefficient real part.
mixed_c0_imag	float	Mixed boundary C0 coefficient imaginary part.
mixed_c1	float	Mixed boundary C1 coefficient real part.
mixed_c1_imag	float	Mixed boundary C1 coefficient imaginary part.
mu_ssd	float	Relative permeability for Small Skin Depth boundary (Mu).
sigma_ssd	float	Conductivity in MS/m for Small Skin Depth boundary (Sig).
inner_angle	float	Inner angle for periodic boundary (degrees).
outer_angle	float	Outer angle for periodic boundary (degrees).

Circuits

Attribute	Type	Description
name	str	Name of the circuit.
total_amps_real	float	Total current (real component) in amps.
total_amps_imag	float	Total current (imaginary component) in amps.
circuit_type	int	Type of circuit.

Methods

Method	Parameters	Description
point_potential	x: float, y: float	Returns A at point (x, y).
field_potential	resolution: int = 300	Returns (x_space, y_space, a_grid).
field_b	resolution: int = 300	Returns (x_space, y_space, bx, by).
point_b	x: float, y: float, eps: float = 1e-6	Returns (bx, by) at point using finite differences.